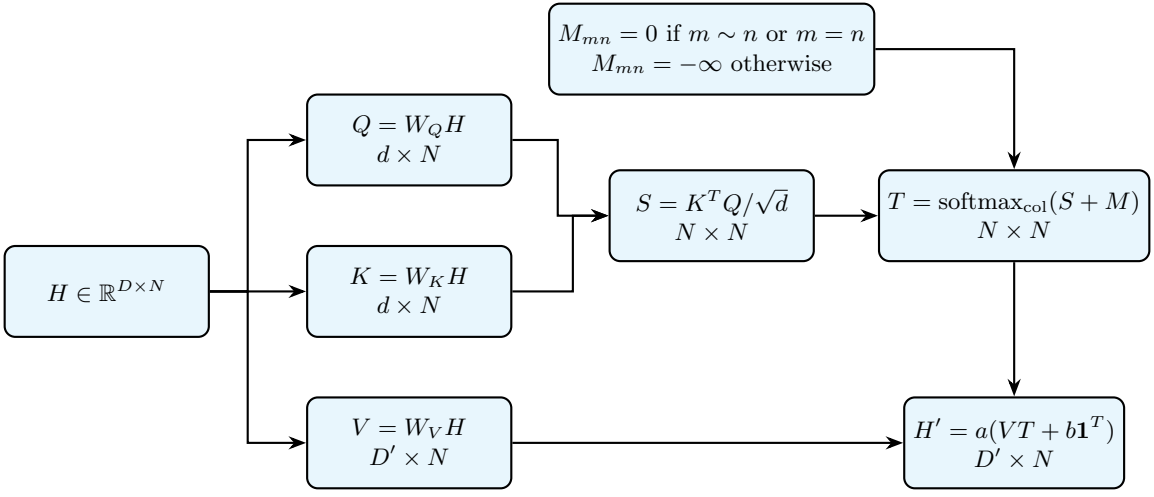

$M_{mn} = 0$ if $m \sim n$ or $m = n$
$M_{mn} = -\infty$ otherwise
$Q = W_Q H$
$d \times N$
$S = K^T Q / \sqrt{d}$
$N \times N$
$T = \text{softmax}_{\text{col}}(S + M)$
$N \times N$
$H \in \mathbb{R}^{D \times N}$
$K = W_K H$
$d \times N$
$V = W_V H$
$D' \times N$
$H' = a(VT + b\mathbf{1}^T)$
$D' \times N$